Spervised Machine Learning

In [ ]:

```
-Labeled data --->Targe Column

Algorithm types:

1)Regression(Continus data)

1)Linear Regression
2)Logistic regression
3)KNearestNeigbourregressor
4)DecisionTreeRegressor
5)RandomForestRegressor
6)AdaboostRegressor


 2)Classification(Discrite data)

1)KNearestNeigbourClassifier
2)DecisionTreeClassifier
3)RandomForestClassifier
4)AdaboostClassifier
```

In [ ]:

In [ ]:

```
step by step solution of problem --->algorithm

Making Tea:

0 1 0 1
Yes No
```

Linear Regression

In [ ]:

```
What is LinearRegression?

Suppose we have have data:

Hourse of studied(X)      Marks(Y)    PREDICTED    ERROR       SE

1                        35             35           0          0
2                         40            41            -1        1
3                        50             49           1          1
4                        55             54           1           1
5                        60             70          -10        100 = MSE 50

X=1   Y=35
X^=3  Y^=50
N=5

We want to answer:

if students studies 6 hours
what marks can predict?


Lr tries to find the best stright line through the data

Mathematical Intution:
y=mx+c  --->Line equation

In machine learning we normally write:

Y^=b0+b1x

where

Y----->predicted output/dependent variable
x --->input/independent/feature
b0---->intercept
b1----->slope/coefficient


Suppose our model learns:

y^=28+7x

For 5 hours:

y^=28+7(10)
y=28+70
y=98
```

In [ ]:

```
What dose the slop mean?

y=28+7x

b1=7
this means:

for every 1 unit increase in study hours
predicted marks increase by approximatly 7 marks

What dose intercept mean?

y^=28+7(0)

b0=28

when x=0 the model predict y=28
y=28

Actual Vs Predicted:

y=65

model prediction:
y^=60

e=y-y^
=65-63


residual/error=2

Why can't we just Draw a any line?

we want the line that gives the smallest overall error

But if we add error:
e1+e2+e3+....

positive and negative errors could cancel each other

(5)^2 + (-5)^2=0

it's look likes there is no error

square the error

Mean Squared Error:

ei=yi-yi^

e^2=(yi - yi^ )^2

Find b0 and b1 that minimize MSE

THIS IS THE HEART OF LR

RSS

How dose the model Find Best Fit Line:

Threre are two approches:

1)OLS (Ordinary least squares):

b1=
b0=?

y=b0+b1x
b0=y-b1x

y=mx+c

y-c/x=m

y-mx=c

2)Gradient Decent
```

In [3]:

```python
import numpy as np

X=np.array([1,2,3,4,5])  #input

Y=np.array([35,40,50,55,65]) #Actual output

#Calculate the mean
x_mean=np.mean(X)
Y_mean=np.mean(Y)

#Calculate slope

numerator=np.sum((X-x_mean) *(Y-Y_mean))

deomininator=np.sum((X-x_mean)**2)

b1=numerator/deomininator

#calculate intercept
b0=Y_mean -b1*x_mean

print("Slope: ",b1)
print("Intercept: ",b0)
```

```
Slope:  7.5
Intercept:  26.5
```

In [5]:

```python
y=26.5+7.5*3
```

In [7]:

```python
y
```

Out[7]:

```
49.0
```

In [9]:

```python
50-49
```

Out[9]:

```
1
```

Calculate MSE Manually

In [12]:

```python
#Predictions for all students

y_pred=b0+b1*X

#Error
error=Y-y_pred
squared_error=error**2

#MSE

mse=np.mean(squared_error)

print("Prediction: ",y_pred)
print("Error: ",error)
print("MSE:",mse)
```

```
Prediction:  [34.  41.5 49.  56.5 64. ]
Error:  [ 1.  -1.5  1.  -1.5  1. ]
MSE: 1.5
```

In [ ]:

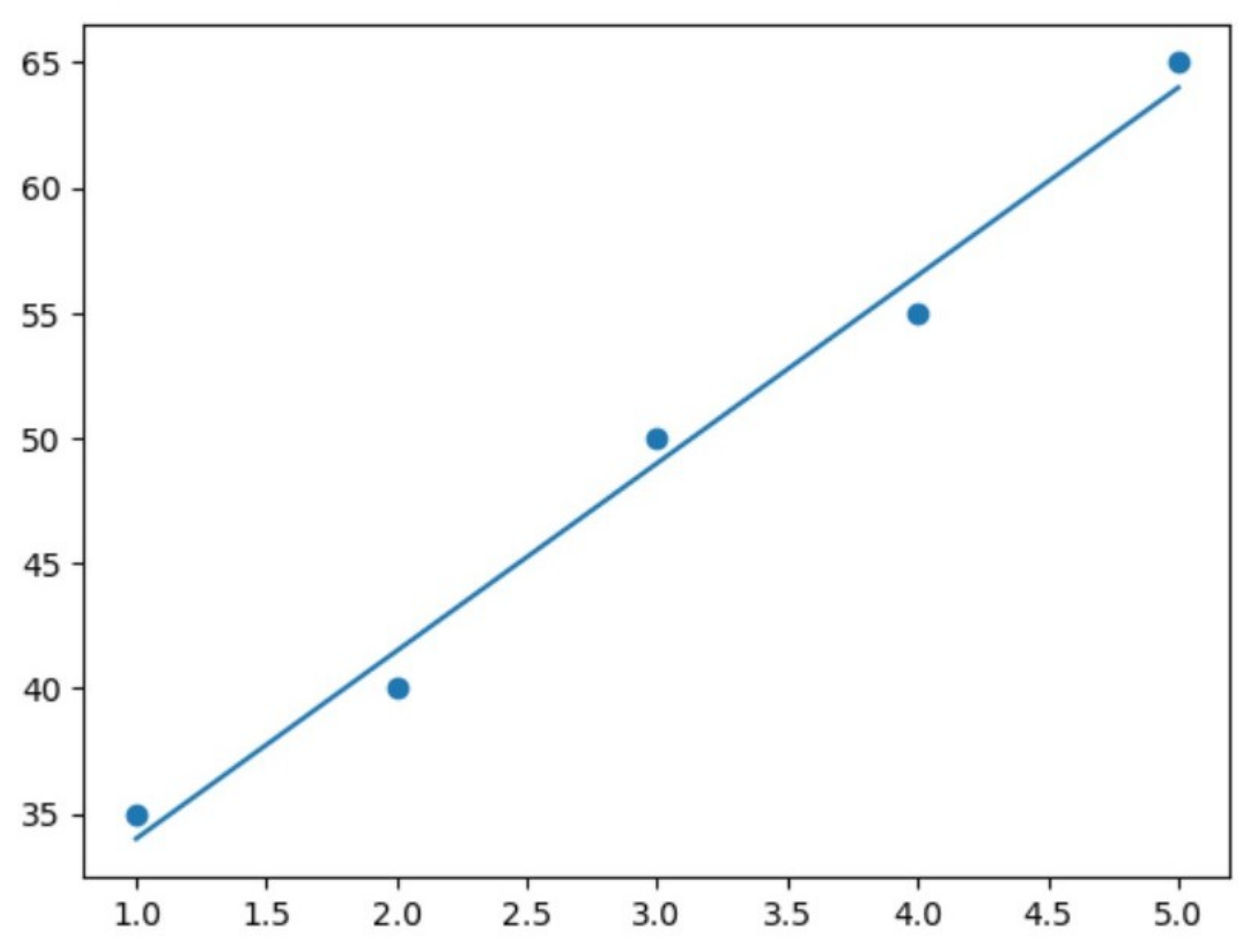
```
Actual -->Prediction --->error --->squared_error -->Average --->MSE
```

In [14]:

```python
import matplotlib.pyplot as plt
plt.scatter(X,Y)
plt.plot(X,y_pred)
```

Out[14]:

```
[<matplotlib.lines.Line2D at 0x1a487f13fe0>]
```



ML Implementation

In [31]:

```python
import  numpy as np
from sklearn.linear_model import LinearRegression

X=np.array([1,2,3,4,5]).reshape(-1,1)
y=np.array([35,40,50,55,65])

model=LinearRegression()

model.fit(X,y)
```

Out[31]:

```
LinearRegression
LinearRegression()
```

In [35]:

```python
model.coef_
```

Out[35]:

```
array([7.5])
```

In [37]:

```python
model.intercept_
```

Out[37]:

```
26.499999999999996
```

In [43]:

```python
new_student=[[50]]

model.predict(new_student)
```

Out[43]:

```
array([401.5])
```

In [ ]: